SANJYOT DAHALE

Unreal Engine Programmer | Mumbai, India (open to remote and relocation) | github.com/LifeisaRepo | sanjyotdahale.dev | sanjyot.code@gmail.com | linkedin.com/in/sanjyot-dahale

Six years in Unreal C++ and Blueprints, from engine plugins for third-party SDKs and native platform APIs to gameplay abilities in GAS and networking. Led the studio's Unreal team; now looking for a hands-on programming role.

WORK EXPERIENCE

LIMINAL XR SOLUTIONS

Lead Unreal Engine Developer & Tech Lead | Jan 2023 - Current

- Led a 7-person team shipping the studio's first published game, Riddler's Ransom, to the **Meta Quest Store**, owning the platform call, performance budgets and certification.
- Planned the PC-VR to standalone Quest 3 port and the optimisation order that took a 30–40 FPS PC prototype to 68–72 FPS on the headset.
- Built an experimental Unreal third-party plugin around Logitech's G29 SDK and exposed it to Blueprint as a callable library.
- Led a Blueprint-to-C++ rebuild of a live product, removing a 10–15% crash rate and raising a runtime feature's ceiling from 25 to 125 instances.
- Wrote the multiplayer control core of Montra Electric's four-player expo racer, turning the host into a non-playing operator console. Ran crash-free for 1,000+ players.
- First Unreal hire at the studio. Built its Unreal team from there, interviewing and mentoring all 10 hires, one of whom now leads the creative team.

LIMINAL XR SOLUTIONS

Unreal Developer & Virtual Production Supervisor | Sep 2020 - Jan 2023

- Designed the camera-alignment system behind a peer-reviewed **SMPTE** paper, reconciling three coordinate systems over standard Unreal replication.
- Integrated camera-tracking, motion-capture and broadcast hardware into Unreal, feeding live transform data into real-time productions.
- Set performance budgets across lighting, shaders and geometry to hold real-time frame targets on live productions.
- Standardised source control studio-wide, establishing GitHub for primary version control and deploying Perforce for specialised projects.

LIMINAL XR SOLUTIONS

Unity Developer | Aug 2019 - Sep 2020

- Built AR and VR applications in Unity, then helped migrate the studio's production pipeline to Unreal Engine.

SKILLS

Languages
C++
Unreal Blueprints
HLSL
Python
C#

Unreal
Gameplay Ability System (GAS)
Behavior Trees & EQS
Animation Blueprints
UMG
Materials & Shaders

Networking
Actor replication
RPCs
Client-server authority

Performance
Performance budgeting
Draw-call & memory optimisation
Mobile performance

Tools
Perforce
Git / Git LFS
Visual Studio

Platforms
PC
Meta Quest 3
Android
HoloLens 2
Meta XR SDK

PROJECTS

IP STREAM MEDIA

Solo Developer · in development (v0.1.2-beta)

An Unreal Engine plugin that plays live RTSP camera streams through Unreal's own **Media Framework**, with FFmpeg decoding on a dedicated worker thread. Reconnects with backoff through standard Media Player events and shuts down cleanly in under 100 ms.

CHRONO RECALL

Solo Developer

A time-rewind ability built in C++ on Unreal's **Gameplay Ability System**. A ring buffer records 3 seconds of movement and health at a fixed 20 Hz; a custom ability task takes over character movement, replays the path with interpolation and restores momentum, then heals the difference through a Set by Caller effect. Locally predicted and tested with two clients.

UE_GEMMA

Solo Developer

Unreal Android build running a 270M function-calling LLM fully on-device and offline, driving five gameplay actions from speech or text. Built a C++ plugin exposing native Android APIs to Blueprint through a **UPL/JNI bridge**, handling async calls across the C++/Java boundary; fine-tuned the model on a script-generated ~3,000-sample dataset.

PUBLICATIONS

IMMERSIVE AND INTERACTIVE AR GRAPHICS AND ENVIRONMENTS FOR BROADCAST APPLICATIONS

SMPTE Motion Imaging Journal · July 2024 · DOI 10.5594/JMI.2024/IHPC9661

CERTIFICATIONS

COMPUTER GRAPHICS WITH MODERN OPENGL AND C++

Udemy | Jan 2025

EDUCATION

UNIVERSITY OF MUMBAI

Bachelor of Engineering, Computer Engineering | 2016 - 2020

LANGUAGES

English	Hindi	Marathi
Full Professional	Professional Working	Native